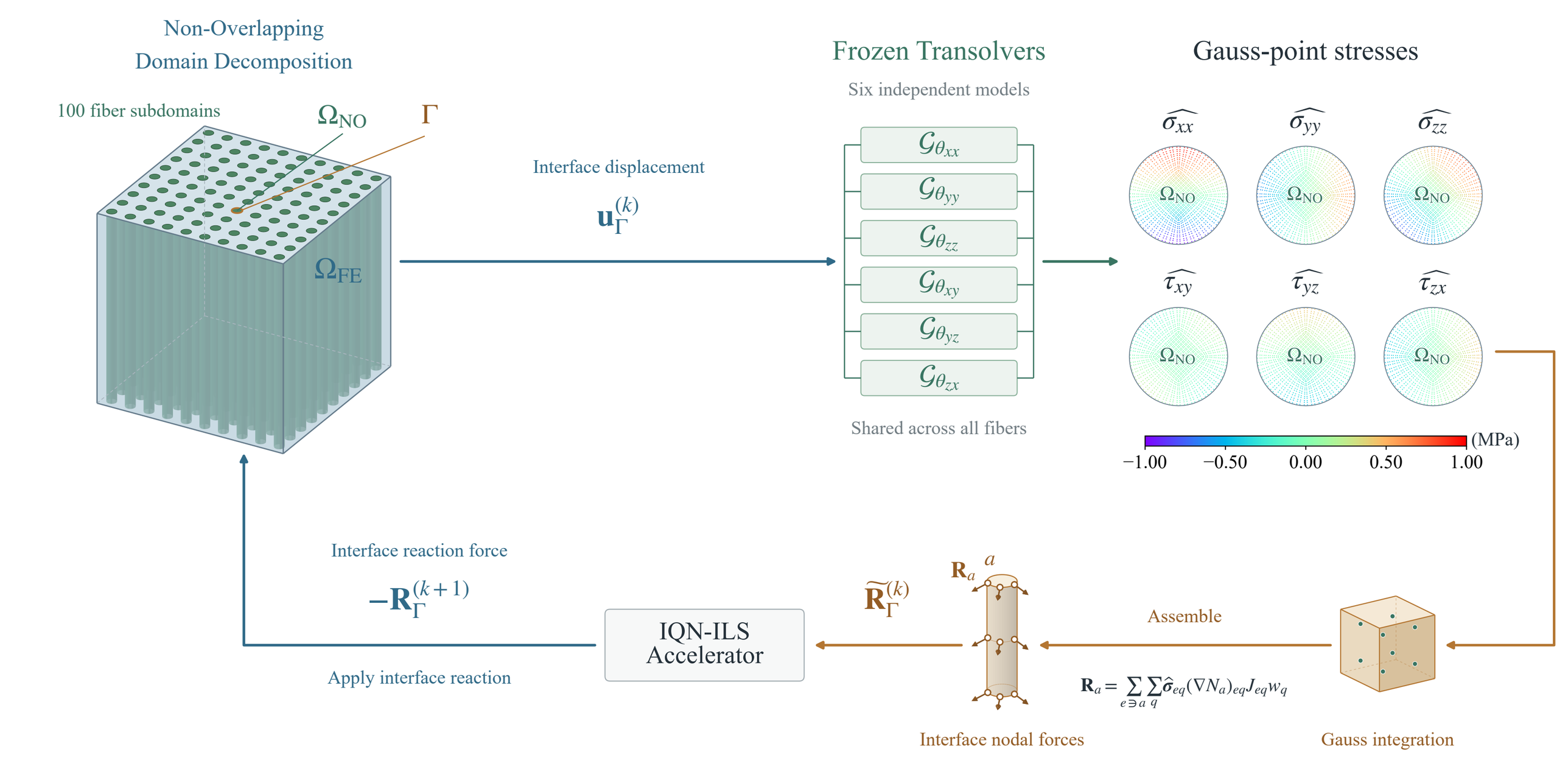

Non-Overlapping
Domain Decomposition
100 fiber subdomains
Ω_{NO}
Γ
Ω_{FE}
Interface displacement
$\mathbf{u}_\Gamma^{(k)}$
Frozen Transolvers
Six independent models
$\mathcal{G}_{\theta_{xx}}$
$\mathcal{G}_{\theta_{yy}}$
$\mathcal{G}_{\theta_{zz}}$
$\mathcal{G}_{\theta_{xy}}$
$\mathcal{G}_{\theta_{yz}}$
$\mathcal{G}_{\theta_{zx}}$
Shared across all fibers
Gauss-point stresses
$\widehat{\sigma_{xx}}$
$\widehat{\sigma_{yy}}$
$\widehat{\sigma_{zz}}$
$\widehat{\tau_{xy}}$
$\widehat{\tau_{yz}}$
$\widehat{\tau_{zx}}$
Ω_{NO}
(MPa)
−1.00
−0.50
0.00
0.50
1.00
Gauss integration
Assemble
$\mathbf{R}_a = \sum_{e \ni a} \sum_q \hat{\boldsymbol{\sigma}}_{eq} (\nabla N_a)_{eq} J_{eq} w_q$
$\mathbf{R}_a$
a
Interface nodal forces
$\widetilde{\mathbf{R}}_\Gamma^{(k)}$
IQN-ILS
Accelerator
Interface reaction force
$-\mathbf{R}_\Gamma^{(k+1)}$
Apply interface reaction